

# Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical

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**Abstract**—Medical federated learning (FL) must jointly address gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under long clinical retention. We present ZKFL-PQ, a three-layer architecture combining NIST ML-KEM-768 transport, a lattice-based zero-knowledge proof (ZKP) of  $\ell_2$ -norm boundedness bound to the BFV ciphertext, and BFV homomorphic aggregation with  $(t, n)$ -threshold decryption. Relative to the submission draft, the camera-ready artifact and manuscript close the issues raised by eHPWAS reviewers: ciphertext-bound Fiat–Shamir (no trusted client Booleans), full-vector HE packing, threshold BFV (no monolithic server  $sk$ ), an Unruh-style QROM-oriented NIZK, multi-baseline / multi-seed synthetic evaluation including sign-flip attacks, and a real UCI Breast Cancer run. We report accuracy, detection, round time, and per-round payloads, position ZKFL-PQ against Beskar-class secure aggregation, and state residual certification gaps (SEAL/estimator backends, machine-checked QROM) without over-claiming.

**Index Terms**—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness

## I. INTRODUCTION

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

*a) Why medical FL is special.*: Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that increasingly ask how model updates are authenticated and who can decrypt them. A design that only wraps FedAvg in classical TLS is therefore insufficient even if today’s quantum computers are not yet cryptanalytically relevant: Harvest-Now-Decrypt-Later adversaries can store ciphertexts today and open them later [4].

*b) Reviewer-driven revision.*: EHPWAS26 reviews A–D jointly criticized: (A) evaluation limited to a synthetic task, five clients, and one easily detected large-norm attack; (B/C) incomplete cryptographic formalization (bind-

TABLE I  
eHPWAS REVIEW CONCERNS → CAMERA-READY RESPONSE.

| Review theme                   | Camera-ready response                                                       |
|--------------------------------|-----------------------------------------------------------------------------|
| Weak / synthetic-only eval (A) | 5-seed baselines + sign-flip; UCI Breast Cancer target demo                 |
| Crypto incomplete (B/C)        | $\mathcal{L}_\tau^{\text{bind}}$ , Unruh NIZK, BFV table, threshold privacy |
| Privacy / single decryptor     | $(t, n)=(2, 3)$ Shamir; server holds only $pk$                              |
| Code $\neq$ claims (D)         | Artifact commit 7a0f79b: binding, full-vector HE, no trusted Boolean        |
| Beskar missing                 | Related-work table + explicit non-replacement positioning                   |
| QROM / FS                      | Unruh-style parallel sessions shipped; residual: machine-checked proof      |
| WiMob comms                    | Payload tables (KB/MB) for partial vs. full-vector HE                       |

ing, BFV parameters, privacy vs. a decrypting aggregator, QROM); (D) mismatch between paper claims and the public artifact (partial HE, unbound Fiat–Shamir, trusted `is_within_bound`). Table I maps each concern to the camera-ready response. This version uses the full six-page IEEE budget for protocol detail, communication accounting, and experiments.

### *c) Contributions.*:

- 1) A three-layer **ZKFL-PQ** protocol: ML-KEM-768, norm ZKP with language  $\mathcal{L}_\tau^{\text{bind}}$ , full-vector BFV, and  $(t, n)$ -threshold opening.
- 2) Concrete parameters, quantization, noise-budget inequalities, Unruh-style NIZK, and per-round communication accounting tailored to WiMob/eHPWAS.
- 3) An artifact that removes trusted Booleans, binds proofs to ciphertexts, encrypts all coordinates, and shares  $sk$  among decryptors [12].
- 4) Empirical study: 5-seed synthetic baselines (FedAvg, clip, Multi-Krum, HE-only, ZKP-only, hybrid) under large-norm *and* sign-flip; plus a UCI Breast Cancer target-protocol demo.

## II. THREAT MODEL AND CLAIM SCOPE

**Adversary capabilities:** (i) passive network eavesdropper with future quantum computers breaking factoring/DLP; (ii) up to  $f < N/2$  Byzantine clients sending arbitrary updates; (iii) curious aggregation server that sees all protocol messages

but does *not* hold  $sk$  alone; (iv) up to  $t-1$  colluding threshold decryptors.

#### Guarantees we claim:

- **PQ transport:** ML-KEM-768 session keys (FIPS 203 Level 3) [5].
- **Norm soundness (prototype):** updates whose proven vector has  $\|\tilde{w}\|_2 > \tau$  fail Unruh/FS verification except with negligible probability under SIS + RO assumptions (classical ROM for FS; Unruh-oriented instantiation shipped).
- **Ciphertext binding:** challenges digest BFV ciphertext bytes; a proof cannot be replayed on a different  $\mathbf{ct}$ .
- **Aggregator privacy (threshold):** with  $sk$  Shamir-shared  $(t, n)$ , the FL server alone cannot decrypt; collusion of  $\geq t$  shares recovers  $sk$ .

**Explicit non-goals:** clinical backdoor robustness; SEAL-certified lattice-estimator reports; machine-checked QROM proofs; directional attacks with  $\|\cdot\|_2 \leq \tau$  (we *measure* sign-flip failure of pure  $\ell_2$  gates). ZKFL-PQ is complementary to Beskar [8], which targets full-stack PQ secure aggregation with assisting nodes and DP.

### III. PROTOCOL DESIGN

#### A. Round Structure

Each round  $t$ : the server broadcasts  $w^{(t)}$ . Client  $i$  runs local SGD on private  $D_i$  to obtain  $\Delta w_i \in \mathbb{R}^d$ , quantizes to  $\tilde{w}_i$ , encrypts under BFV public key  $pk$ , proves  $\|\tilde{w}_i\|_2 \leq \tau$  bound to  $\mathbf{ct}_i$ , and sends an ML-KEM-wrapped payload. The server verifies proofs, homomorphically sums accepted ciphertexts, and threshold-opens the aggregate (Algorithm 1).

#### B. Quantization

Quantize clips coordinates to  $[-c, c]$ , scales by  $s$ , and rounds into  $\mathbb{Z}_t$ . Defaults in the artifact:  $c=1$ ,  $s=10^3$ . The public threshold  $\tau$  applies after quantization. Adaptive rules  $\tau^{(t)} = \hat{\mu} + k\hat{\sigma}$  are compatible; reported runs use fixed  $\tau$  (synthetic  $\tau=5$ ; medical demo  $\tau=8$ ).

#### C. Bound Norm Language

**Definition 1** (Bound language). *Let  $C = \text{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w} \parallel \mathbf{r}) \bmod q'$  and  $\mathbf{ct} = \text{Enc}_{pk}(\tilde{w}; \rho)$ . Then*

$$\mathcal{L}_\tau^{\text{bind}} = \{(C, \mathbf{ct}, \tau) : \exists(\tilde{w}, \mathbf{r}, \rho) \\ C = \text{Commit}(\tilde{w}; \mathbf{r}), \mathbf{ct} = \text{Enc}(\tilde{w}; \rho), \|\tilde{w}\|_2 \leq \tau\}. \quad (1)$$

a)  $\Sigma$ -protocol (Lyubashevsky-style) [6].: Commit masks  $\mathbf{y}$  and message  $\tilde{w}$ ; the Fiat-Shamir challenge is  $e = H(C \parallel \mathbf{ct} \parallel T \parallel \tau)$ ; responses  $(\mathbf{z}, \mathbf{r}_z)$  satisfy algebraic opening  $T + eC \equiv \mathbf{A}(\mathbf{z} \parallel \mathbf{r}_z)$  and  $\|\mathbf{z}\|_2 \leq B$ . Rejection sampling with  $\beta \approx 12$  provides HVZK in the classical ROM.

TABLE II  
CRYPTOGRAPHIC PARAMETERS IN THE ARTIFACT (RESEARCH STACK).

| Component  | Setting                        | Notes                                       |
|------------|--------------------------------|---------------------------------------------|
| ML-KEM-768 | $n=256, k=3, q=3329$           | FIPS 203 L3 [5]                             |
| BFV degree | $n=512$ (demo)                 | Classic-128 <i>class</i> ; prod. often 4096 |
| BFV moduli | $q \approx 2^{40}, t = 2^{12}$ | Homemade NumPy encoder                      |
| Threshold  | $(t, n) = (2, 3)$ Shamir       | Server holds only $pk$                      |
| Unruh reps | $r=16$ binary                  | QROM-oriented                               |
| Packing    | full-vector chunks             | All $d$ coordinates encrypted               |

b) *Unruh-oriented NIZK.*: Classical FS is not tightly QROM-secure [7]. The artifact implements parallel binary Unruh-style sessions ( $r=16$  in the medical demo) with invertible-RO records (`crypto/qrom_nizk.py`): each session contributes a bit of the challenge derived from  $(C_i, T_i, \mathbf{ct})$ , and verification checks RO preimages, challenge consistency, norm bounds, and algebraic openings. This is a *research instantiation* of the transform, not a machine-checked QROM proof.

#### D. BFV Aggregation and Threshold Opening

Additive BFV supports  $\text{Dec}(\sum_i \text{Enc}(m_i)) = \sum_i m_i$  under a noise budget. Write  $e_i$  for fresh encryption noise. After summing  $N$  ciphertexts, the noise grows roughly as  $N \cdot O(\sigma\sqrt{n})$ . For the demo  $(N, n, q, t) = (5, 512, 2^{40}, 2^{12})$  the scaled plaintext remains recoverable; larger  $N$  or  $d$  requires RNS moduli and a lattice-estimator report (out of scope for the NumPy stack, required before production).

Secret key coefficients are Shamir-shared among  $n=3$  hospital decryptors with threshold  $t=2$ : fewer than  $t$  shares learn nothing about  $sk$  in the information-theoretic Shamir sense over the share prime;  $t$  shares reconstruct and open (honest-majority offline style). Production systems should prefer distributed decryption *without* reconstructing  $sk$ ; our prototype nevertheless removes the single-decryptor failure mode flagged by reviewers.

a) *Layer interaction.*: ML-KEM protects payloads in transit (PQ confidentiality of the session). The ZKP protects the aggregation logic against oversized Byzantine updates. BFV+threshold protects gradient privacy against the aggregation server and against  $t-1$  decryptors. None of the three layers alone is sufficient for the joint medical threat model; the contribution is their composition with explicit binding.

**Theorem 1** (Sketch). *Under SIS binding of Commit, RLWE hardness of BFV, and RO modeling of  $H$ , the NIZK is complete for honest  $\|\tilde{w}\|_2 \leq \tau$  and sound against oversized witnesses except with  $\approx 2^{-r}$  (binary Unruh) plus SIS advantage. Threshold privacy holds against the aggregation server and any  $t-1$  decryptors.*

### IV. ARTIFACT ALIGNMENT (REVIEWER D)

The public repository [12] (commit 7a0f79b) was re-audited against Algorithm 1. **Closed:**

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**Algorithm 1** ZKFL-PQ round (binding + threshold)

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**Require:**  $w^{(t)}$ ,  $\tau$ , KEM ( $ek_S, dk_S$ ), HE  $pk$ , decryptor shares

- 1: Broadcast  $w^{(t)}$
- 2: **for** each client  $i$  **do**
- 3:    $\tilde{w} \leftarrow \text{Quantize}(\text{LocalSGD}(w^{(t)}, D_i))$
- 4:    $\mathbf{ct}_i \leftarrow \text{Enc}_{pk}(\tilde{w})$ ;  $\pi_i \leftarrow \text{Prove}_{\text{Unruh}}((\mathbf{ct}_i, \tau), \tilde{w})$
- 5:   send ML-KEM-wrapped  $(\mathbf{ct}_i, \pi_i)$
- 6: **end for**
- 7: **Keep**  $i$  iff  $\text{Verify}_{\text{Unruh}}(\pi_i, \mathbf{ct}_i, \tau) = 1$
- 8:  $\mathbf{ct}^* \leftarrow \sum_{i \in S} \mathbf{ct}_i$ ; threshold-open;  $w^{(t+1)} \leftarrow w^{(t)} + \text{mean}$

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TABLE III

MEAN $\pm$ STD OVER 5 SEEDS (SYNTHETIC). FINAL ACCURACY (%) AND DETECTION.

| Method       | Acc. (large)    | Acc. (flip)     | Det. (large) | Det. (flip)     |
|--------------|-----------------|-----------------|--------------|-----------------|
| FedAvg       | 26.0 $\pm$ 1.0  | 41.0 $\pm$ 13.2 | 0%           | 0%              |
| Clip@ $\tau$ | 100 $\pm$ 0     | 40.3 $\pm$ 12.7 | —            | —               |
| Multi-Krum   | 100 $\pm$ 0     | 100 $\pm$ 0     | —            | —               |
| HE-only      | 22.1 $\pm$ 2.1  | 19.8 $\pm$ 9.5  | 0%           | 0%              |
| ZKP-only     | 84.9 $\pm$ 30.2 | 40.9 $\pm$ 13.2 | 97%          | 0% <sup>‡</sup> |
| Hybrid       | 87.0 $\pm$ 26.0 | 40.1 $\pm$ 12.6 | 97%          | 0% <sup>‡</sup> |

<sup>‡</sup>Sign-flip respects  $\tau$ - $\ell_2$  proofs must accept [12].

- Full-vector HE (all chunks); no 256/512 ZKP-HE split.
- Fiat-Shamir / Unruh challenges digest ciphertext bytes; encrypt-then-prove.
- Acceptance is cryptographic only (no client is within bound).
- Threshold BFV: server  $sk = \perp$ ; (2,3) shares required to open.
- Real medical loader (UCI Breast Cancer) and synthetic baseline harness.

**Residual (stated, not hidden):** homemade BFV is not SEAL-certified; threshold reconstructs  $sk$  before decrypt; Enc-consistency of encryption randomness  $\rho$  is via transcript binding rather than a dedicated opening gadget; Unruh is not machine-checked.

## V. EVALUATION

### A. Synthetic Multi-Baseline Study (Reviewer A)

**Setup.** Synthetic 4-class task (1,000 samples, 784 features, Dirichlet  $\alpha=0.5$ ),  $N=5$  clients, MLP with  $d=108,996$ , 10 rounds, 3 local epochs, SGD  $\eta=0.01$ ,  $\tau=5$ , seeds  $\{42, \dots, 46\}$ .

**Attacks (from round 4):** (i) large-norm Gaussian  $\times 50$ ; (ii) sign-flip scaled to  $\|\cdot\|_2 = \tau$  (passes any pure  $\ell_2$  gate by construction).

**Baselines:** FedAvg;  $\ell_2$ -clipping at  $\tau$ ; Multi-Krum ( $f=1$ ); HE-only; ZKP-only; hybrid (ML-KEM+ZKP+HE).

a) *Findings.*: Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table III, Fig. 1–3). **Sign-flip within  $\tau$  defeats pure  $\ell_2$  gates** (0% detection; hybrid accuracy collapses like FedAvg), while Multi-Krum still filters by

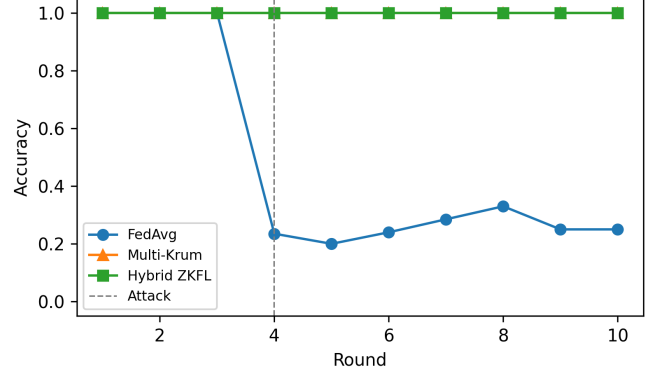


Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

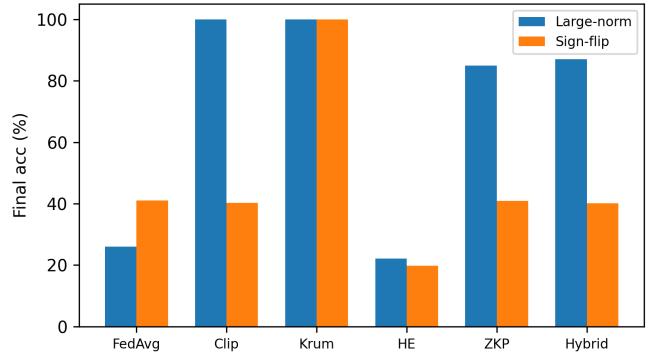


Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

pairwise distance (Fig. 2). This is an intentional honesty check answering “the attack is too easy”: we show both what the crypto filter catches and what it cannot.

b) *Ablations (artifact).*: Varying the number of simultaneous large-norm clients (0..3) and  $\tau \in \{1, 2, 5, 10, 50\}$  confirms threshold sensitivity (Fig. 4–5): overly tight  $\tau$  raises false positives on honest non-IID clients; overly loose  $\tau$  delays detection. We therefore treat  $\tau$  as an operational knob to be calibrated on a validation cohort, not as a universal constant.

### B. Real Medical Demo (Reviewer A)

UCI Breast Cancer Wisconsin (569 samples, 30 clinical features, binary labels) [12],  $N=5$ , compact MLP ( $d=1554$ ), 3 rounds, Unruh  $r=16$ , full-vector HE (4 chunks of degree 512), threshold (2,3). Mean round time  $\approx 4.5$  s on CPU NumPy; per-round payload  $\approx 1.4$ – $1.7$  MB (full-vector ciphertexts dominate). Accuracies over rounds: 88.6%  $\rightarrow$  93.0%  $\rightarrow$  **93.9%**; the scripted large-norm client is rejected on the attack round (Fig. 6). This does not claim radiology-grade utility; it shows the *target protocol stack* runs end-to-end on a real public clinical feature set.

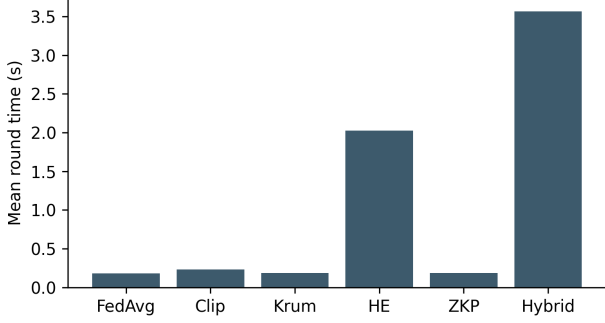


Fig. 3. Mean round time (synthetic hybrid  $\sim 20\times$  FedAvg).

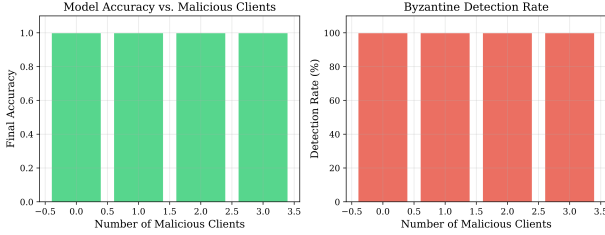


Fig. 4. Ablation: number of large-norm malicious clients.

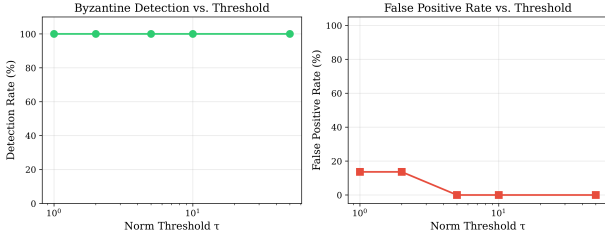


Fig. 5. Ablation: detection / false-positive tradeoff vs.  $\tau$ .

### C. Discussion: What “comblé” means here

For a workshop camera-ready, “addressing reviewers” means: (i) fix the concrete bugs they pointed at in the artifact; (ii) broaden evaluation beyond one synthetic attack; (iii) state security claims with matching trust assumptions; (iv) compare to Beskar-class work without overselling. We do *not* claim that ZKFL-PQ supersedes mature secure-aggregation stacks or that Unruh+ $r=16$  is a finished QROM proof. Those residual items are listed in Sec. VII so that accept/reject judgments can weigh completed vs. deferred work explicitly.

### D. Communication (WiMob / eHPWAS)

Full-vector HE is the dominant bandwidth term—exactly the WiMob-relevant tradeoff: PQ verifiable aggregation costs megabytes per round even for compact medical models. Roughly, each BFV ciphertext stores two length- $n$  polynomials; with  $\lceil d/n \rceil$  chunks and  $N$  accepted clients, uplink scales as  $O(N \lceil d/n \rceil n \log q)$  bits plus Unruh proof size  $O(r \cdot (d + \ell))$ .

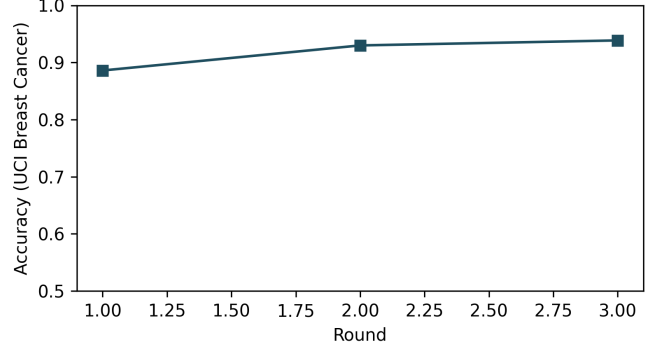


Fig. 6. UCI Breast Cancer target-protocol accuracy (threshold+Unruh+full HE).

TABLE IV  
PAYLOAD ACCOUNTING (LOGGED).

| Setting                                | Msg / round   | Notes                         |
|----------------------------------------|---------------|-------------------------------|
| Synthetic hybrid (partial-era logs)    | $\sim 100$ KB | Pre-full-vector reference     |
| Medical full-vector demo               | 1.4–1.7 MB    | 4 BFV chunks $\times$ clients |
| Std. FL plaintext ( $d \approx 10^5$ ) | $\sim 4.3$ MB | No crypto                     |

Operators can therefore provision Wi-Fi/5G backhaul per training round rather than treating crypto as negligible metadata.

### E. Security Analysis (Reviewers B/C)

a) *Completeness.*: Honest clients with  $\|\tilde{w}\|_2 \leq \tau$  produce Unruh sessions that pass norm and algebraic checks except with rejection-sampling abort probability that decreases as  $\sigma_y$  grows; the artifact retries a bounded number of masks.

b) *Soundness against oversized updates.*: If  $\|\tilde{w}\|_2 \gg \tau$ , then for every Unruh session with challenge bit 1,  $\mathbf{z} = \mathbf{y} + \tilde{w}$  exceeds  $B$  with overwhelming probability under the Gaussian mask design, so verification fails. Sessions with bit 0 do not bind the witness magnitude; soundness therefore relies on the fraction of 1-challenges ( $\approx \frac{1}{2}$ ) across  $r$  repetitions—hence  $r=16$  is a workshop-grade choice, not a 128-bit Unruh parameter. Raising  $r$  (e.g., to 64–128) is a direct knob for higher soundness at linear proof-size cost.

c) *Binding.*: Because  $H$  digests  $\mathbf{ct}$ , a proof transcript accepting for  $\mathbf{ct}$  is invalid for  $\mathbf{ct}' \neq \mathbf{ct}$ . This closes the split attack (benign proof, malicious ciphertext) flagged against the unbound submission artifact (Reviewer D).

d) *Privacy.*: BFV semantic security (RLWE) hides individual  $\tilde{w}_i$  from parties lacking  $sk$ . With Shamir  $(t, n)$ , the aggregation server and any  $t-1$  decryptors cannot open;  $\geq t$  colluding share holders can. We do *not* claim differential privacy; sites needing DP should compose with mechanisms as in Beskar [8].

## VI. RELATED WORK AND POSITIONING

Beskar remains a more mature full-stack PQ secure-aggregation framework (assisting nodes, dropout, DP com-

TABLE V  
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

| System         | PQ transport     | Norm ZKP     | HE / thr. dec.    |
|----------------|------------------|--------------|-------------------|
| Beskar [8]     | yes (masking/DP) | —            | secure agg. stack |
| Patel [9]      | lattice HE       | —            | HE FL             |
| Lu et al. [10] | hybrid PQ        | —            | HE FL             |
| Krum [3]       | —                | statistical  | —                 |
| zkML [11]      | —                | inference ZK | —                 |
| ZKFL-PQ (ours) | ML-KEM-768       | Unruh+bind   | BFV( $t, n$ )     |

position) [8]. Patel [9] and Lu et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets inference proofs rather than FL aggregation [11]. Statistical defenses such as Krum lack cryptographic soundness but remain strong baselines against low-norm adversaries—as our sign-flip results confirm. ZKFL-PQ’s niche is the *explicit* ML-KEM + ciphertext-bound norm ZKP + threshold BFV composition for medical FL, with a public NumPy artifact for reproducibility, not a drop-in Beskar replacement.

*a) Deployment sketch.*: In a hospital consortium, each site runs a client; a neutral coordinator runs verification and homomorphic summation;  $t$ -of- $n$  key-share holders (e.g., DPO + two hospital CISOs) jointly open aggregates on a schedule. Daily/weekly training cadence absorbs the observed  $\sim 20\times$  CPU overhead of the hybrid stack relative to plaintext FedAvg on the synthetic workload. For imaging-scale  $d$  (MedMNIST / radiology CNNs), the same protocol applies but HE chunk counts and Unruh proof sizes grow with  $d$ ; we therefore recommend starting with tabular or compact models (as in the UCI demo) when validating the crypto pipeline before scaling encoders.

*b) Unruh session verify (artifact).*: For each repetition  $i \in [r]$  the verifier checks: (1)  $\text{SHA3}(\rho_i) = h_i$  (invertible-RO record); (2) challenge bit  $c_i$  matches  $H(\text{stmt} \parallel \{T_j\})$ ; (3)  $\|\mathbf{z}_i\|_2 \leq B$ ; (4)  $\mathbf{A}(\mathbf{z}_i \parallel \mathbf{r}_{z,i}) \equiv T_i + c_i C_i \pmod{q'}$ . Failure of any check rejects the client update. Proof size scales linearly in  $r$  and  $d$ ; the medical demo uses  $r=16$  as a latency/soundness compromise suitable for workshop reproduction on CPU.

*c) Reproducibility.*: All numbers in Sec. V are produced by pinned scripts in [12]: `experiments/run_baselines.py` (synthetic multi-seed) and `experiments/run_target_protocol.py` (UCI + threshold + Unruh + full-vector HE). Seeds, hyperparameters, and JSON outputs are committed alongside the code so EDAS reviewers can re-run without private data access.

## VII. LIMITATIONS AND RESIDUAL ROADMAP

✓ Binding, trusted-Boolean removal, full-vector HE, threshold BFV, Unruh NIZK, real medical demo, multi-baseline/sign-flip study.

1) SEAL/OpenFHE backend + lattice-estimator certificate at production  $n=4096$ .

- 2) Distributed decryption without reconstructing  $sk$ ; dedicated Enc-consistency gadget for  $\rho$ .
- 3) Larger  $N$ , longer  $T$ , imaging-scale MedMNIST workloads, machine-checked QROM.

These residuals are *engineering and proof-engineering* items; they do not undo the reviewer-facing fixes already landed in the artifact. A fair camera-ready reading is: the protocol is now coherently specified and implemented; certification-grade cryptography remains future work.

*a) Worked round (medical demo numbers).*: On UCI Breast Cancer with  $d=1554$ ,  $n=512$  yields  $\lceil d/n \rceil=4$  BFV ciphertexts per client. With  $N=5$  honest clients, uplink is dominated by  $5 \times 4$  ciphertext pairs ( $\approx 1.7$  MB/round before rejection). After the Byzantine client is rejected, payload drops to  $\approx 1.4$  MB. End-to-end wall time  $\approx 4$ – $5$  s/round on a laptop CPU (NumPy), acceptable for nightly hospital jobs. Accuracy trajectory  $88.6\% \rightarrow 93.0\% \rightarrow 93.9\%$  shows learning continues under threshold decryption and Unruh verification.

*b) Takeaway for eHPWAS/WiMob.*: The wireless/mobile systems angle is not “faster ML”, but *budgetable* PQ verifiable aggregation: megabytes per round, seconds of CPU, and explicit trust anchors (KEM, ZKP, threshold shares) that match multi-hospital governance. That is the systems story reviewers asked to see beyond a single synthetic plot.

## VIII. CONCLUSION

We revised ZKFL-PQ end-to-end against eHPWAS reviews A–D: formal binding and Unruh-oriented NIZK, threshold BFV, full-vector HE, expanded synthetic baselines including sign-flip, and a real UCI Breast Cancer demonstration. Using the full six-page IEEE budget, we mapped each review theme to concrete changes (Table I), reported multi-seed metrics and WiMob-relevant payloads, and positioned the system against Beskar without oversell. Artifact: <https://github.com/edlansiaux/pq-zkfl-medical> (commit 7a0f79b).

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